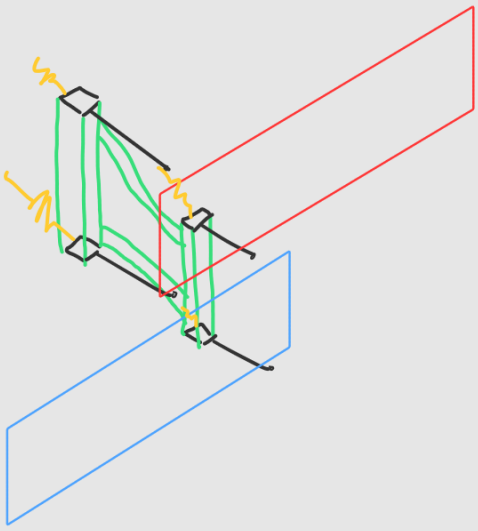
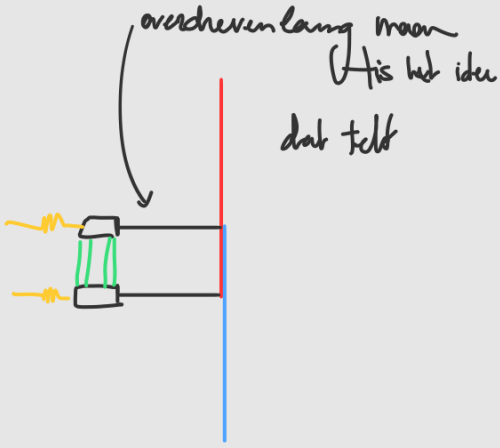
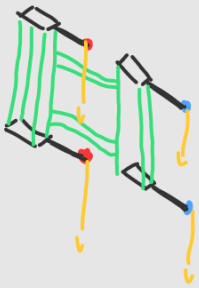
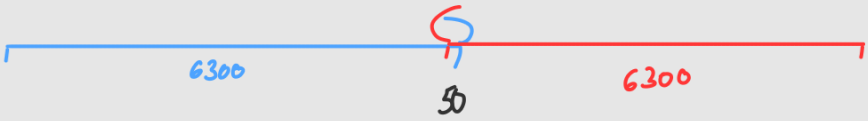


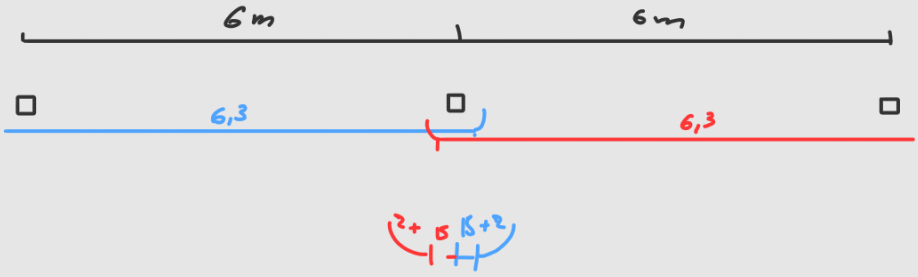


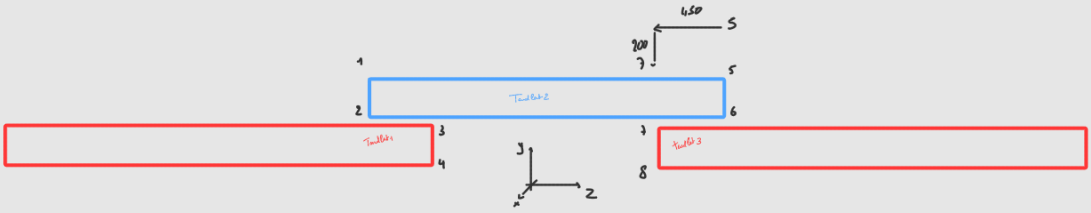


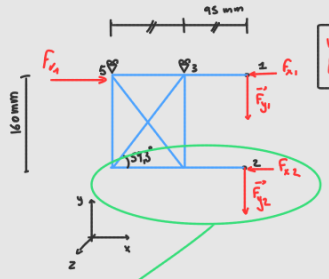
Statische benadering.

$\downarrow = \vec{G}_{\text{tondlat sectie}}/4$









WAT TE BEPALEN: alle interne krachten & momenten in onbekende vgl. vorm	oplossen via onbekende om zo ins Code te kunnen stoken

"Aannames":
- eigengewicht verwaarlozen
- Alles is aan elkaar gelast
- constructie draait stil
- alle zol opleggingen zijn glygoits
  ↳ $\pm F_y$ kunnen ze opvangen
  ↳ kunnen momenten opvangen

↳ beschouw als uitkeling
Het buigt zichtbaar door
voor een balk van 30 30

↳ $I = \frac{30 \cdot 30^3}{12}$

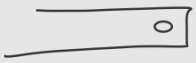
$$v_{max} = \frac{-PL^3}{3EI}$$

$$v_{max} = \frac{-P \cdot 95\,mm^3}{3 \cdot 210000\,N/mm^2 \cdot \frac{30^4}{12}\,mm^4}$$

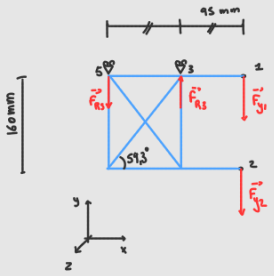
$$= 0{,}000020162\,mm/N \cdot P$$

voor 1mm doorbuiging moet
dit al een P krachten van 49,6 kN ⟶ dus meer dan 5 ton

klinkt overdreven
maar hier kunnen wel een gat in van d=10mm



Moment zonder s:



$$\Sigma \vec{M}_s = 0 = +F_{R3} \cdot 0{,}095\,m - (F_{y1} + F_{y2}) \cdot (2 \cdot 0{,}095\,m)$$

$$\rightsquigarrow F_{R3} = \frac{(F_{y1} + F_{y2}) \cdot 2 \cdot \cancel{0{,}095\,m}}{\cancel{0{,}095\,m}}$$

$$F_{R3} = 2 \cdot (F_{y1} + F_{y2}) \qquad (1)$$

Krachten evenwicht $\Sigma \vec{F}_y = 0$:

$$\Sigma F_y = 0$$

$$\Rightarrow \vec{F}_{R5} + \vec{F}_{R3} + \vec{F}_{y1} + \vec{F}_{y2} = 0$$

$$\vec{F}_{R5} - \left(2 \cdot (\vec{F}_{y1} + \vec{F}_{y2})\right) + \vec{F}_{y1} + \vec{F}_{y2} = 0 \qquad (1)$$

$$\vec{F}_{R5} - (\vec{F}_{y1} + \vec{F}_{y2}) = 0$$

$$\vec{F}_{R5} = \vec{F}_{y1} + \vec{F}_{y2} \qquad (2)$$